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BioMAC-Sim-Toolbox: A MATLAB toolbox for biomechanical motion analysis and creation through simulation

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Summary

We present a MATLAB toolbox for human movement simulation and analysis. The toolbox is versatile due to its object-oriented design, such that different dynamics models can be easily combined with different problem classes. The toolbox's focus is on creating simulations of human walking and running by solving trajectory optimization problems. The trajectory optimization problems are solved using direct collocation. In these problems, the muscle stimulations and initial state are found that minimize an objective related to energy expenditure. Different objectives have been implemented that represent energy or effort minimization. We have also included different objectives for data tracking, including tracking of joint angles, ground reaction forces, accelerometer data, gyroscope data, and marker positions. We have implemented two different two dimensional (2D) musucloskeletal models and one three dimensional (3D) musculoskeletal model. To show how the toolbox can be used, we include a 2D and a 3D introductory example, as well as different applications that are based on previous publications. In the future, we aim to implement different other problem classes as well, such as inverse kinematics and dynamics to process experimental data, static and dynamic optimizations to find muscle simulations from joint moments, while also forward shooting can be used to investigate neural control of gait.

Statement of Need

Movements of humans and other animals are extremely versatile, while our efficiency is unmatched by human-made machines without having accurate or efficient controllers. Therefore, a better understanding of human movement can have a large impact for persons with movement disabilities, sports performance optimization, and design of human-made devices. Movement simulations are a key tool for creating this understanding. On the one hand, we can use so-called predictive simulations (Ackermann & Van den Bogert, 2010; Falisse et al., 2019; Koelewijn et al., 2018) to investigate unseen movements by replicating the optimization in the central nervous system (Zarrugh et al., 1974) in a computer optimization. On the other hand, we can use so-called reconstructive simulations to estimate variables that cannot or were not measured directly by minimizing a data tracking error (Dorschky, Nitschke, et al., 2019; Nitschke et al., 2023, 2024). Such wide-spread measurements and their biomechanical analysis are vital to be able to reach this understanding of human movement.

Here, we present `BioMAC-Sim-Toolbox`, a MATLAB toolbox that can be used to create simulations of human movement and analyse human movements. The main functionality of

the toolbox is that it solves trajectory optimization problems, or optimal control problems, for human musculoskeletal dynamics models. These dynamics models combine multibody dynamics to model the movements of the skeleton with muscle dynamics models to model the dynamics of muscular contraction and activation. In the trajectory optimization problem, the muscle stimulations are found that minimize an objective. This objective generally includes terms to minimize muscular effort and to minimize a tracking error with respect to measured movement data (Dorschky, Nitschke, et al., 2019; Dorschky, Krüger, et al., 2019; Koelewijn & Bogert, 2016; Koelewijn & Selinger, 2022; Nitschke et al., 2023, 2024), but different other objectives have been implemented as well, such as minimization of metabolic energy expenditure (Koelewijn et al., 2018). The optimization problem can be described mathematically as follows:

$$\underset{\mathbf{x}(0),\mathbf{u}(t),v,T}{\text{minimize}} \quad J(\mathbf{x},\mathbf{u}) = \sum_{j=1}^{N_W} W_j \int_{t=0}^{T} c_j(x(t),u(t))\mathrm{d}t + c_f(x(T),u(T))$$

$$\text{subject to} \quad \mathbf{f}(\mathbf{x}(t),\dot{\mathbf{x}}(t),\mathbf{u}(t)) = \mathbf{0} \text{ for } 0 \leq t \leq T \text{(system dynamics)}$$

$$\mathbf{x_L} \leq \mathbf{x}(t) \leq \mathbf{x_U} \text{ for } 0 \leq t \leq T \quad \text{(bounds on states)}$$

$$\mathbf{u_L} \leq \mathbf{u}(t) \leq \mathbf{u_U} \text{ for } 0 \leq t \leq T \quad \text{(bounds on controls)}$$

for a musculoskeletal model with state x and input u. Here, the goal is to find the initial state, $\mathbf{x}(0)$, control inputs, $\mathbf{u}(t)$, the speed, v, and duration, T, that minimize the cost function, $J(\mathbf{x},\mathbf{u})$. The cost function consists of $c_j(\mathbf{x}(t),\mathbf{u}(t))$, which describes the cost functions that are evaluated over time, which are multiplied with associated weigth W_j, and of a final cost function $c_f(\mathbf{x}(T),\mathbf{u}(T))$, which is only evaluated at the end of the trajectory, at time T. We normally do not include a final cost function. The subscript L defines the lower bounds, while U defines the upper bounds. When speed or duration is known, we commonly still include them into the optimization variables, while making the lower and upper bound equal to the desired value.

For walking and running, we include a periodicity constraint to the full gait cycle or a single step, in case of a symmetric motion. This constraint can be defined for straight and curved running and walking (Nitschke et al., 2020) through the displacement in the horizontal plane. This periodicity constraint ensures that the states are the same at the end of the trajectory as at the beginning, while applying a horizontal displacement to the states in $\mathbf{x}_{hor}$:

$$\mathbf{x}(T) = \mathbf{x}(0) + vT\mathbf{x}_{hor} \quad \text{(periodicity)}$$

So far, we have created the problem class 'Collocation', which creates a trajectory optimization problem using direct collocation (Betts, 2010). In direct collocation, the numerical integration of the differential equations that represent the dynamics are solved as equality constraints during the optimization. To do so, collocation nodes, or time nodes, need to be predefined, and the integration method should be selected. We have currently implemented a backward Euler and a midpoint Euler integration method. Backward Euler has shown to be most stable and has been used for publications. A direct collocation approach creates a large-scale nonlinear optimization problem, for which the derivatives of the objectives and all constraint can be derived analytically. An optimization problem including a periodicity constraint could be transcribed as follows after applying direct collocation with backward Euler integration:

$$\underset{\mathbf{x}(i),\mathbf{u}(i),v,T}{\text{minimize}} \quad J(\mathbf{x},\mathbf{u}) = \sum_{j=1}^{N_W} W_j \sum_{i=0}^{N} c_j(\mathbf{x}(i),\mathbf{u}(i)) + c_f(\mathbf{x}(N),\mathbf{u}(N))$$

$$\text{subject to} \quad \mathbf{f}(\mathbf{x}(i+1), \frac{\mathbf{x}(i+1)-\mathbf{x}(i)}{\Delta t}, \mathbf{u}(i)) = \mathbf{0} \text{ for } i = 1, \dots, N \quad \text{(system dynamics)}$$

$$\mathbf{x_L} \leq \mathbf{x}(i) \leq \mathbf{x_U} \text{ for } i = 1, \dots, N \quad \text{(bounds on states)}$$

$$\mathbf{u_L} \leq \mathbf{u}(i) \leq \mathbf{u_U} \text{ for } i = 1, \dots, N \quad \text{(bounds on controls)}$$

$$\mathbf{x}(N+1) = \mathbf{x}(0) + vT\mathbf{x}_{hor} \quad \text{(periodicity)}$$

for N collocation points i.

Problems can be solved using the solvers that are implemented in the class 'solver'. So far, we have implemented IPOPT (Wächter & Biegler, 2006) to solve the resulting optimization problem. This algorithm can solve optimal control problems transcribed with direct collocation in less than 10 minutes for a 2D model (Koelewijn & Bogert, 2016; Koelewijn & Selinger, 2022) and less than one hour for a 3D model (Nitschke et al., 2020). Different solvers could easily be integrated into this solver class.

As most analysis is specific to the problem type, most code for analysis can be found in the respective problem class. For example, metabolic cost requires an integration over time, which is dependent on the problem type that is being used. In addition, general methods, e.g., to calculate a correlation or root mean squared error between two variables, or to write results into an OpenSim file format, can be found in the function 'HelperFunctions' folder.

We have implemented different human dynamics models. All implemented models are musculoskeletal models, but they can also be used as skeletal models, such that the input is generated using joint moments instead of muscle stimulations. We have implemented a 3D model version (gait3d) and two 2D model versions in c, which are compiled as mex functions. The model dynamics can be tested using the test cases coded in the 'tests' folder. The 3D model and one 2D model (gait2d_osim) are loaded from OpenSim, but use our own muscle dynamics model, which is described in (Nitschke et al., 2020), while the other (gait2dc) is used in our own previous work (Dorschky, Nitschke, et al., 2019; Dorschky, Krüger, et al., 2019; Koelewijn & Bogert, 2016; Koelewijn & Selinger, 2022). The model parameters are defined in the .osim file for the OpenSim models, and defined in an Excel file for model gait2dc (gait2dc_par.xlsx). The models can be personalized by directly adjusting the parameters in the .osim model or Excel file, such that, for example, OpenSim scaling can be used (Seth et al., 2011). They can also still be adjusted in MATLAB, for example to investigate virtual participants (Dorschky, Nitschke, et al., 2019; Koelewijn & Selinger, 2022). The model dynamics are explained further for gait2dc (Dorschky, Nitschke, et al., 2019; Dorschky, Krüger, et al., 2019; Koelewijn & Selinger, 2022), for gait3d (Nitschke et al., 2020). The gait2d_osim model has not yet been used in publications. It is based on the gait10dof18musc.osim model, and can be loaded in two ways: the original version, called gait10dof18musc, and a version with the lumbar joint locked, called gait2d_osim.

We have included several examples in the folder 'ExampleScripts' to highlight the applications of the model and help future users get started with the code. We have added two introduction examples, one for 2D simulations (script2D.m) and one for 3D simulations (script3D), which is based on Nitschke et al. (2020). The goal of these introductory examples is to show how the toolbox works, and highlight different options in the implementation. We have added an application in the folder 'Treadmill' to show an example use of the 'gait2d_osim' model, combined with an implementation of a treadmill. Based on previous publications, we have added three additional applications. The code in the folder 'IMU2D' shows how simulations can be created that reconstruct inertial sensor movement, and is based on the

publications by Dorschky, Nitschke, et al. (2019) and Dorschky et al. (2024). The code in the folder 'ExoPaper' shows how simulations can be used to make predictions, in this case of walking with an exoskeleton, based on the paper by Koelewijn & Selinger (2022). The code in 'MarkerTracking3D' is an example of how simulations can be created by tracking marker and ground reaction force data, based on the publication by Nitschke et al. (2024).

Currently, the toolbox can be used to solve trajectory optimization problems to solve predictive and reconstructive simulations. We have implemented code to track marker positions, joint angles, translations, ground reaction forces, accelerations, angular velocities, movement duration, and movement speed. We have further implemented objectives to minimize muscular effort, metabolic cost, squared torque, joint accelerations, passive joint moments, ground reaction force impact, and head stability. By combining these different objectives, many different types of simulations can be generated and the movement kinematics and kinetics investigated.

In future, the toolbox can be expanded. New problem classes can be defined, for example for inverse kinematics and inverse dynamics, or to solve muscle activations and stimulations statically or dynamically. Furthermore, a single shooting approach can be implemented, which would allow for models with discontinuous dynamics or control to be investigated, such as a reflex model (Geyer & Herr, 2010).

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